AI

Develop Expert System

Name: Arjun Mankuskar

PRN:202401110038

Batch:A2

Introduction

Expert Systems mimic the decision-making of human technicians by applying a set of rules to a knowledge base of facts. In the **Automobile** domain, they can rapidly diagnose vehicle issues from sensor warnings and driver-reported "symptoms." This report presents a simple rule-based expert system in Python that takes car "symptoms" as input and suggests likely faults.

Knowledge Base

Facts (Possible Car Symptoms)

1. Engine light
2. knocking noise
3. low oil pressure
4. over heating
5. battery warning

6. clicking noise
7. brake warning
8. ABS light
9. Tire pressure low
10. high mileage

Rules (IF–THEN format)

1. IF {engine_light, knocking_noise} THEN Engine Knock
2. IF {low_oil_pressure, high_mileage} THEN Oil Wear
3. IF {battery_warning, clicking_noise} THEN Dead Battery
4. IF {overheating, engine_light} THEN Overheating Engine
5. IF {brake_warning, ABS_light} THEN Brake System Fault
6. IF {tire_pressure_low, high_mileage} THEN Tire Tread Wear

7. IF {engine_light, low_oil_pressure} THEN Oil Leak
8. IF {overheating, low_oil_pressure} THEN Coolant Leak
9. IF {clicking_noise, brake_warning} THEN Brake Pump Issue
10. IF {battery_warning, engine_light} THEN Charging System Issue

#Code

```python
def diagnose(symptoms):
    """
    symptoms: set of strings representing car warnings/symptoms
    returns: list of matching fault diagnoses
    """
    rules = [
        ({"engine_light", "knocking_noise"},   "Engine Knock"),
        ({"low_oil_pressure", "high_mileage"}, "Oil Wear"),
        ({"battery_warning", "clicking_noise"},"Dead Battery"),
        ({"overheating", "engine_light"},      "Overheating Engine"),
        ({"brake_warning", "ABS_light"},       "Brake System Fault"),
        ({"tire_pressure_low", "high_mileage"},"Tire Tread Wear"),
        ({"engine_light", "low_oil_pressure"}, "Oil Leak"),
        ({"overheating", "low_oil_pressure"},  "Coolant Leak"),
        ({"clicking_noise", "brake_warning"},  "Brake Pump Issue"),
        ({"battery_warning", "engine_light"},  "Charging System Issue"),
    ]

    matches = []
    for condition, fault in rules:
        if condition.issubset(symptoms):
            matches.append(fault)
    return matches or ["No fault detected."]


if __name__ == "__main__":
    print("Welcome to the Automobile Fault-Diagnosis Expert System\n")
    raw = input("Enter observed car symptoms (comma separated): ").lower()
    # normalize to keys
    car_symptoms = {
        s.strip().replace(" ", "_")
        for s in raw.split(",")
        if s.strip()
    }

    results = diagnose(car_symptoms)
    print("\nPossible Diagnoses based on your input:")
    for r in results:
        print("-", r)
```

Sample Run & Output

1.

```
PS D:\VS code\AI phython>  python automobile_expert_system.py
Welcome to the Automobile Fault-Diagnosis Expert System

Enter observed car symptoms (comma separated): engine light, knocking noise

Possible Diagnoses based on your input:
- Engine Knock
PS D:\VS code\AI phython>
```

2.

```
PS D:\VS code\AI phython> python automobile_expert_system.py
Welcome to the Automobile Fault-Diagnosis Expert System

Enter observed car symptoms (comma separated): low oil pressure, high mileage

Possible Diagnoses based on your input:
- Oil Wear
PS D:\VS code\AI phython>
```

3.

```
PS D:\VS code\AI phython>  python automobile_expert_system.py
Welcome to the Automobile Fault-Diagnosis Expert System

Enter observed car symptoms (comma separated): battery warning, clicking noise, engine light

Possible Diagnoses based on your input:
- Dead Battery
- Charging System Issue
PS D:\VS code\AI phython>
```

4.

```
PS D:\VS code\AI phython> python automobile_expert_system.py
Welcome to the Automobile Fault-Diagnosis Expert System

Enter observed car symptoms (comma separated): engine light, low oil pressure

Possible Diagnoses based on your input:
- Oil Leak
PS D:\VS code\AI phython>
```

5.

```
PS D:\VS code\AI phython> python automobile_expert_system.py
Welcome to the Automobile Fault-Diagnosis Expert System

Enter observed car symptoms (comma separated): overheating, low oil pressure

Possible Diagnoses based on your input:
- Coolant Leak
PS D:\VS code\AI phython>
```

6.

```
PS D:\VS code\AI phython> python automobile_expert_system.py
Welcome to the Automobile Fault-Diagnosis Expert System

Enter observed car symptoms (comma separated): brake warning, ABS light

Possible Diagnoses based on your input:
- No fault detected.
PS D:\VS code\AI phython>
```

7.

```
PS D:\VS code\AI phython> python automobile_expert_system.py
Welcome to the Automobile Fault-Diagnosis Expert System

Enter observed car symptoms (comma separated): clicking noise, brake warning

Possible Diagnoses based on your input:
- Brake Pump Issue
PS D:\VS code\AI phython>
```

Conclusion

This simple expert system demonstrates how rule-based reasoning can automate vehicle fault diagnosis from basic symptom inputs. It highlights the importance of a well-structured knowledge base and clear IF–THEN rules.

Future Enhancements

- Integrate real sensor data (e.g., OBD-II readings)
- Add confidence scores for overlapping rules
- Support numeric thresholds instead of only Boolean “symptoms”
- Build a GUI or web interface for user-friendly interaction